

Heart Disease Risk Prediction

Owais Jawaid

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Predicting Heart Disease Risk Using Machine Learning Techniques on Clinical Patient Data

Introduction

Cardiovascular diseases (CVDs) remain the leading cause of death globally, accounting for millions of deaths every year. In fact, an “estimated 19.8 million people died from CVDs in 2022, representing approximately 32% of all global deaths” (World Health Organization, 2025). Early detection of heart disease can dramatically improve patient outcomes by enabling timely clinical intervention. However, traditional diagnostic methods often rely on expensive tests or subjective evaluation, making early detection inconsistent and sometimes inaccessible.

Machine learning (ML) offers a powerful alternative. By analyzing multiple clinical features simultaneously, ML can detect patterns associated with heart disease risk that may be difficult for humans to identify. These computational models can complement clinical judgment, reduce false negatives, and offer fast and cost-effective screening tools.

The primary objective of this project is to build and evaluate a machine learning model capable of predicting heart disease using routine clinical data. Multiple classification algorithms, including Logistic Regression, Naive Bayes, Random Forest, Support Vector Machine (SVM), and XGBoost, were trained and compared. Their performance was evaluated based on accuracy, precision, recall, F1-score, and ROC-AUC. The project ultimately identifies the best-performing model and highlights the clinical variables most strongly associated with heart disease.

Dataset Description

The dataset used for this project is the *Heart Disease Dataset* obtained from Kaggle (heart.csv). It includes common clinical features typically gathered during medical examinations. Before cleaning, the dataset contained 1025 rows; however, many duplicates were identified. After removing them, 302 unique patient records remained. No missing values were found, allowing preprocessing to proceed smoothly.

Dataset Structure:

- **Total rows (after cleaning):** 302
- **Number of features:** 13 clinical features + 1 target variable

Target Variable: target – Binary variable:

The target variable indicates the presence of heart disease:

- 1 = heart disease present
- 0 = no heart disease

Distribution:

- 164 patients with disease
- 138 patients without disease

Feature Overview:

Features include:

- **age:** Age of the patient (in years)
- **sex:** Gender (0 = female, 1 = male)
- **cp:** Chest pain type (0 – 3)
 - 0 = typical angina
 - 1 = atypical angina
 - 2 = non-anginal pain
 - 3 = asymptomatic
- **trestbps:** Resting blood pressure
- **chol:** Serum cholesterol (mg/dl)
- **fbs:** Fasting blood sugar > 120 mg/dl (1 = true, 0 = false)

- **restecg:** Resting electrocardiographic results:
 - 0 = normal
 - 1 = ST-T wave abnormality
 - 2 = left ventricular hypertrophy
- **thalach:** Maximum heart rate achieved
- **exang:** Exercise-induced angina (1 = yes, 0 = no)
- **oldpeak:** ST depression induced by exercise
- **slope:** Slope of the peak exercise ST segment:
 - 0 = upsloping
 - 1 = flat
 - 2 = downsloping
- **ca:** Number of major vessels (0 – 3) colored by fluoroscopy
- **thal:** Thalassemia (blood disorder type):
 - 1 = Normal
 - 2 = Fixed defect
 - 3 = Reversible defect

These features represent a mix of physiological measurements and diagnostic indicators that are commonly associated with cardiovascular risk.

Methodology

The methodology for this project consisted of five major phases: data cleaning, preprocessing and feature engineering, exploratory data analysis (EDA), model training, and performance evaluation.

Data Cleaning:

- Removed all duplicate rows, reducing dataset from 1025 to 302 rows.
- Verified that no missing values existed.
- Checked target distribution to ensure balanced representation.

Preprocessing & Feature Engineering:

- Converted categorical variables (cp, thal, slope, sex, fbs, etc.) to categorical types.
- Created an engineered feature: age_group (Young, Middle-Aged, Senior, Elderly) for additional interpretability.
- Applied one-hot encoding to all categorical features.
- Standardized numerical columns (chol, trestbps, thalach, oldpeak, age) using StandardScaler.
- Split dataset into 80% training (241 rows) and 20% testing (61 rows).

Exploratory Data Analysis (EDA):

EDA included:

- Histograms and boxplots to examine feature distributions
- Correlation heatmap to identify relationships between features
- Count plots for categorical variables
- Distribution plots comparing healthy vs diseased patients

Key findings:

- Chest pain type (cp) is strongly associated with heart disease.
- Patients with heart disease had lower max heart rates (thalach).
- Higher ST depression values (oldpeak) were linked with disease.
- Exang and ca also showed meaningful differentiation.

Model Selection & Training (Classification Problem):

The problem is a binary classification task (presence or absence of heart disease). 5 models were trained and evaluated:

- **Logistic Regression**
 - Linear baseline model
 - Provides interpretable coefficients
- **Naïve Bayes**
 - Simple probabilistic classifier
 - Useful benchmark for comparison

- **Random Forest Classifier**
 - Ensemble of decision trees
 - Handles feature interactions well
- **Support Vector Machine (SVM)**
 - Effective for small-to-medium datasets
 - Captures non-linear relationships (linear kernel used after tuning)
- **XGBoost**
 - Gradient boosting algorithm
 - Strong performance in structured datasets

Hyperparameter Tuning:

Used GridSearchCV (5-fold cross-validation) for:

- Random Forest
- SVM
- XGBoost

Best hyperparameters found:

- **Random Forest:** max_depth = 8, n_estimators = 100
- **SVM:** C = 1, kernel = 'linear'
- **XGBoost:** learning_rate = 0.2, max_depth = 3, n_estimators = 100

Evaluation Metrics:

The following classification metrics were used to evaluate model performance:

- **Accuracy** – Overall correctness of the model
- **Precision & Recall** – Focus on minimizing false negatives
- **F1-Score** – Balance between precision and recall
- **ROC-AUC Score** – Ability of the classifier to distinguish between classes
- **Confusion Matrix** – Detailed error breakdown

These metrics help evaluate model performance, especially important in healthcare to reduce false negatives.

Result Analysis

Model Performance Comparison:

Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC
Logistic Regression	0.852	0.875	0.848	0.862	0.899
Naive Bayes	0.820	0.844	0.818	0.831	0.911
Random Forest	0.787	0.833	0.758	0.794	0.887
XGBoost	0.787	0.833	0.758	0.794	0.851
SVM	0.869	0.879	0.879	0.879	0.910

Best Model:

The Support Vector Machine (SVM) model achieved:

- **Highest accuracy:** 86.9%
- **High precision and recall:** 87.9% each
- **Strong ROC-AUC:** 0.91

This made SVM the most balanced and reliable model overall.

ROC Analysis:

- SVM and Naive Bayes both achieved high ROC-AUC (~0.91)
- Strong separation indicates good diagnostic ability

Confusion Matrix (SVM):

Correct predictions: **53**

Incorrect predictions: **8**

This reflects strong performance on unseen data.

Feature Importance:

Using XGBoost for interpretability, the top features were:

1. **cp** (Chest pain type)
2. **thalach** (Max heart rate)
3. **oldpeak** (ST depression)
4. **exang** (Exercise-induced angina)
5. **ca** (# of major vessels)

These findings align with known medical indicators of cardiovascular risk.

Conclusion

This project successfully developed and evaluated machine learning models for predicting heart disease using routine clinical data. The preprocessing and feature engineering steps ensured clean, consistent inputs. Through comparison of multiple models, the Support Vector Machine (SVM) model emerged as the best performer, achieving the highest accuracy and strong classification metrics across the board.

The project also identified the most important clinical predictors, including chest pain type, maximum heart rate, ST depression, exercise-induced angina, and number of major vessels. These indicators are consistent with medical literature and further validate the approach.

Machine learning has significant potential as a decision-support tool in healthcare. With additional data, deployment in a real-world clinical environment, and integration with patient-facing tools (such as risk calculators), this system could support earlier detection and improve patient outcomes.

References

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